

FS-PREM: A Physics-Aware Framework for Predicting Port Disruption

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Abstract—The US Gulf Coast contains some of the busiest and globally important ports in the world, serving as critical gateways for petroleum, natural gas, and agricultural exports. However, these ports face immense vulnerability to storm-induced disruptions, costing them billions of dollars and having severe economic consequences. Existing storm risk models focus narrowly on surge prediction, depend on pre-COVID economic assumptions, or lack uncertainty calibration, limiting their operational effectiveness today. To address this issue, this paper introduces FS-PREM, Few-Shot Port Risk Event Mining, a novel framework for predicting port disruption events using real-time hurricane advisories, even when data is extremely scarce. FS-PREM integrates a physics-aware PORTEX index capturing Hazard, Proximity, Uncertainty, Exposure, and Vulnerability dimensions with streaming conformal prediction to provide calibrated uncertainty estimates. The model was evaluated on 120,000+ advisory-port observations across five major Gulf Coast ports: Houston, New Orleans, Mobile, Tampa, and Corpus Christi, and results demonstrate that the physics-aware weighted PORTEX model achieved AUC 0.833 which is an 11 percent improvement over baselines while maintaining 99+ percent uncertainty coverage. Coefficient analysis reveals that Vulnerability (+17.61) and Hazard (+3.21) are primary risk drivers, validating that inadequate infrastructure and storm intensity are the most important indicators in risk prediction. Furthermore, lightweight sequential models, such as Compact GRU with an AUC 0.797, capture temporal dependencies for real-time deployment. This framework addresses critical gaps in port risk assessment, providing a robust foundation for operational decision-making under extreme weather uncertainty.

I. INTRODUCTION

A. Problem Definition

The U.S. Gulf Coast is home to some of the nation’s busiest ports, including Houston, New Orleans, Mobile, Tampa, and Corpus Christi. These ports are critical gateways for petroleum, liquefied natural gas, and agricultural exports, yet they are highly vulnerable to storm-driven disruptions. Specifically, in the aftermath of COVID-19, U.S. Gulf Coast ports have faced compounded vulnerabilities from both supply chain fragility and extreme weather [1]. In August 2021, the U.S. Coast Guard suspended vessel movements at the Port of New Orleans and the whole lower Mississippi due to Hurricane Ida, leading to days-long delays in cargo throughput. Moreover, precautionary measures at Houston and Mobile created ripple effects across energy, grain, and chemical exports. Unlike pre-COVID conditions, port closures now interact with labor

shortages, congested supply chains, and heightened market volatility, amplifying disruption costs. Yet despite their operational importance, port-specific risk events remain difficult to predict from hurricane advisories alone, especially under data scarcity where only a handful of post-COVID storms exist.

B. Current Approaches

Looking at the existing literature in the field, Davila Hernandez et al. [2] applied machine-learning techniques to forecast hurricane storm surges in the Lower Laguna Madre. Though their model demonstrated the feasibility of data-driven surge forecasting, it has a narrow focus on hydrodynamics and did not extend to operational disruptions at ports. Similarly, Balakrishnan et al. [3] proposed a methodology to quantify the economic risks of hurricane-induced port closures. However, their approach was based on pre-COVID economic data and did not account for storm heterogeneity or the real-time nature of advisory updates.

More recent works have explored advanced machine learning approaches for rare-event prediction. Ian et al. [4] utilized bidirectional LSTM networks to assess storm surge risk and capture temporal dependencies in extreme climate events. However, their model lacked streaming calibration, making it less effective for real-time operational decisions.

Taken together, these findings demonstrate the promise which advanced machine learning models have in tackling storm risk prediction. However, existing models are either surge-centric, reliant on pre-COVID datasets, or overlook the uncertainty calibration needed for actionable early warnings, revealing critical gaps in existing systems.

C. Proposed Approach

To address these limitations, this paper introduces the Few-Shot Port Risk Event Mining (FS-PREM) benchmark, constructed exclusively from 2023–2024 Gulf Coast storm data. In contrast to pre-COVID methods that treat advisories as static inputs, FS-PREM integrates physics-aware storm signals, streaming conformal prediction, and lightweight sequential architectures into a unified framework. By structuring advisory data into five interpretable PORTEX dimensions, Hazard, Proximity, Uncertainty, Exposure, and Vulnerability, FS-PREM provides a physics-grounded foundation for modeling port-level risk. Building on this foundation, the framework

applies rolling-window conformal prediction to generate calibrated uncertainty bands with each new advisory, and compact recurrent and convolutional models to capture evolving storm trajectories. This layered approach directly addresses the few-shot setting in which historical storms cannot be assumed representative of current operating conditions.

D. Results Overview

FS-PREM significantly outperforms conventional approaches for rare port disruption prediction. The physics-aware weighted PORTEX model achieved an AUC of 0.833, which represents an 11 percent improvement over baseline classifiers that struggled with the extreme class imbalance (5 percent positive events). Coefficient analysis reveals that Vulnerability (+17.61) and Hazard (+3.21) dominate risk prediction, while Proximity exhibits a counterintuitive negative coefficient (-0.62), suggesting storms closer to ports follow more predictable trajectories.

Streaming conformal prediction maintained predictive performance (AUC 0.74–0.75) while also providing statistically valid uncertainty coverage of 99.4% for $\alpha = 0.05$ and 99.2% for $\alpha = 0.10$. Furthermore, lightweight sequential models further enhanced temporal modeling, with Compact GRU achieving AUC 0.797 (+6.3% over baselines) and Compact Conv1D reaching 0.757, with both of these models maintaining computational efficiency suitable for real-time deployment on streaming advisory data.

II. BACKGROUND

A. Rare event Forecasting in Port Disruption

Figure 1 depicts that hurricane-induced port disruptions are inherently rare events, with only around five percent of advisory–port instances exceeding the 95th percentile disruption threshold in FS-PREM. Each instance is represented as $(x_{p,t}, y_{p,t})$, where $x_{p,t}$ encodes port–storm features under advisory t and $y_{p,t} \in \{0, 1\}$ indicates a disruption. This extreme class imbalance necessitates the model to be capable of learning from limited positive crisis events while also providing calibrated uncertainty estimates, which are critical for operational use. Standard supervised learning assumes sufficient positive examples for reliable pattern recognition. However, rare-event scenarios violate this assumption, leading to models that achieve high accuracy through trivial negative prediction while failing to capture critical positive cases. This motivates the integration of physics-informed features and uncertainty quantification to compensate for data scarcity through domain knowledge.

B. PORTEX Disruption Index

Conventional indices, such as the CME Hurricane Index (CMEHI) [5] and the Hurricane Severity Index (HSI)[6], measure storm hazard primarily using wind speed and storm size. While this approach is effective for insurance and energy markets, these indices are storm-centric and fail to incorporate port-specific exposure and operational sensitivity. Furthermore, wind-only scales, such as Saffir–Simpson [7],

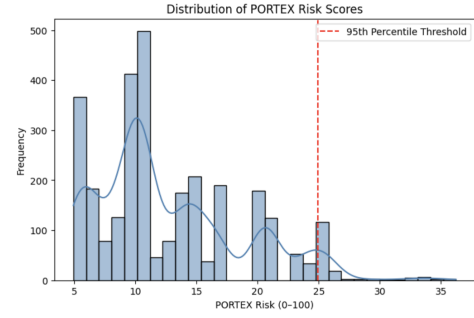


Fig. 1. Class Imbalance Across Ports and Storm Events

ignore surge, rainfall, and logistical disruptions, rendering them unsuitable for this analysis.

To address the limitations, this paper introduces PORTEX, a port-level disruption index that integrates hurricane hazard with operational and infrastructural risk. PORTEX for port p at time t is computed as a weighted combination of five components:

$$\text{PORTEX}_{p,t} = 100 \left[0.5 (H_{p,t} \cdot P_{p,t}) + 0.2 U_{\text{norm}} + 0.2 E_{p,t} + 0.1 V_{p,t} \right]$$

where the hazard term $H_{p,t}$ combines sustained wind speed V and gale radius R as:

$$H_{p,t} = \left(\frac{V}{74} \right)^3 + \frac{2}{3} \left(\frac{R}{60} \right) \left(\frac{V}{74} \right)^2$$

and the proximity term $P_{p,t}$ modulates the hazard by distance to the port:

$$P_{p,t} = \max \left(0, 1 - \frac{d_{p,t}}{1.5 R_{\text{km}}} \right)$$

Forecast uncertainty U_{norm} captures normalized advisory errors, while operational exposure $E_{p,t}$ and vulnerability $V_{p,t}$ quantify port-specific traffic, throughput efficiency, and infrastructural fragility.

PORTEX integrates the strengths of existing indices by accounting for wind intensity like the Saffir–Simpson scale, incorporating storm size like the Hurricane Severity Index, and integrating forecast uncertainty and impact weighting similar to the CMEHI. Furthermore, unlike traditional storm-centric indices, PORTEX provides port-level resolution by gating hazards by proximity and integrating both forecast uncertainty and operational stress, providing a more robust and practical measure of disruption.

C. Sequential Forecasting Under Advisory Streams

FS-PREM is a sequential prediction problem, meaning that as storm advisories arrive, risk estimates must be updated dynamically. For a storm with T advisories, the feature sequence $\{x_{p,1}, \dots, x_{p,T}\}$ produces predictions:

$$\hat{y}_{p,t} = f_{\theta}(x_{p,1:t})$$

where f_{θ} may be linear, ensemble-based, or sequential neural models (e.g., GRU or Conv1D). This streaming formulation

enables continuous risk assessment as storms evolve, making it particularly well suited for operational decision making.

D. Conformal Prediction for Calibrated Uncertainty

Given the rare-event and streaming nature of FS-PREM, calibrated uncertainty is critical. Conformal prediction generates sets $C_{p,t}$ that satisfy:

$$\mathbb{P}(y_{p,t} \in C_{p,t}) \geq 1 - \alpha$$

where α is the tolerated miscoverage, ensuring distribution-free, valid uncertainty estimates, even under inter-storm distributional shifts.

III. METHODS

A. Data Collection

The FS-PREM benchmark is built by integrating storm advisories from the International Best Track Archive for Climate Stewardship (IBTrACS) [8] and vessel movement records from MarineCadastre.gov [9]. The merged dataset covers more than 120,000 advisory–port observations across five of the largest US Gulf Coast ports: Houston, New Orleans, Mobile, Tampa, and Corpus Christi. Each observation corresponds to the interaction between a six-hour hurricane advisory and contemporaneous port-level traffic conditions. Advisories were aligned with AIS records through timestamp matching, ensuring that storm dynamics and port traffic states were consistent.

B. Preprocessing

To achieve port-centric representation, we recomputed the five PORTEX dimensions—Hazard (H), Proximity (P), Uncertainty (U), Exposure (E), and Vulnerability (V)—for every advisory–port pair. This port-centric representation was preferred as each observation reflects how a storm uniquely impacts individual ports rather than treating all affected locations uniformly. Hazard integrates sustained wind speed and gale radius into a nonlinear intensity proxy. Proximity modulates hazard by minimum distance from storm center to the port, while Uncertainty normalizes forecast error in storm trajectory. Exposure aggregates port-level operational features derived from traffic throughput, vessel classes, and passenger flows. Vulnerability quantifies operational fragility by comparing current traffic loads against port efficiency baselines. All features were recomputed directly from IBTrACS and AIS inputs to remove missing values and ensure comparability across storms.

Risk events were defined using percentile-based thresholds on the PORTEX distribution. A ninety-fifth percentile cutoff produced approximately five percent positives, yielding an extreme rare-event classification setting. To assess robustness, alternative thresholds at the ninetieth and ninety-seventh percentiles were also analyzed, producing a sensitivity table that highlights the tradeoff between event rarity and dataset balance.

C. Model Architecture

First, standard classifiers, logistic regression, random forests [10] and gradient-boosted trees [11], were trained on the engineered signals, and then evaluation followed a leave-one-storm-out (LOSO) cross-validation strategy, in which all advisories from a given storm were withheld during training in order to directly test model resilience under storm-to-storm heterogeneity, a key property of few-shot learning. Performance was reported using AUC, PR-AUC, precision, recall, F1, and confusion matrices.

To incorporate domain knowledge, a weighted linear model was formulated over PORTEX components, where coefficients were estimated with sparsity-inducing L1 and group lasso penalties. This method was utilized to preserve interpretability by constraining weights to physical dimensions while allowing the data to re-weight hazard, proximity, and operational exposure according to empirical risk patterns.

Because storm advisories arrive sequentially, a rolling-window conformal prediction wrapper was adopted for both logistic regression and the weighted PORTEX model. At each advisory update, the conformal procedure outputs calibrated prediction sets with nominal 95 percent coverage. This approach provides statistically valid uncertainty quantification under distributional shifts between storms, ensuring predictions remain actionable in real time.

Finally, to capture temporal dependencies in storm evolution, a lightweight recurrent and convolutional model was obtained. Compact GRU [12] and Conv1D [13] architectures with hidden dimensions below sixteen were applied to PORTEX sequences, reflecting the operational constraint of fast inference. Training again followed LOSO validation, and both predictive performance and runtime per advisory were reported to assess feasibility for streaming deployment.

IV. RESULTS

A. Baseline Performance and Rare-Event Challenges

Initial evaluation of standard classifiers, specifically Logistic Regression, Random Forest, Support Vector Machines, and Gradient Boosting, demonstrated that conventional learning algorithms struggle under the extreme class imbalance inherent in FS-PREM. Leave-One-Storm-Out (LOSO) validation was employed in order to simulate deployment conditions where each storm presents novel dynamics, revealing substantial overfitting. For instance, Gradient Boosting exhibited near-perfect performance on training data ($\text{AUC} \approx 0.99$) but declined to 0.73 on held-out storms, evidencing a pronounced generalization gap.

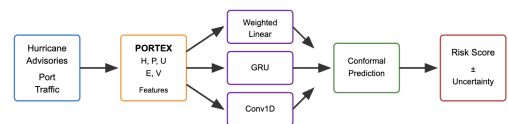


Fig. 2. Pipeline architecture for the proposed system

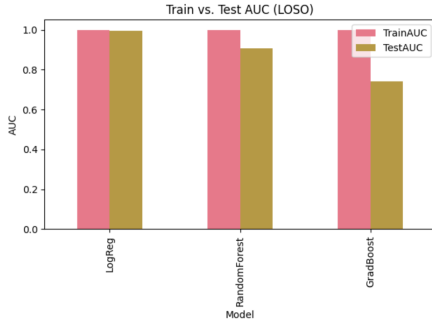


Fig. 3. Baseline Train versus Test AUC

Confusion matrices further emphasize operational limitations. Despite high overall accuracy ($> 94\%$), Logistic Regression failed to detect 60% of high-risk events, and Random Forest exceeded an 80% miss rate. Per-storm AUC varied from near-perfect (1.0) to near-random (≈ 0.5), indicating that each storm behaves as a distinct domain. These findings highlight that naive application of standard classifiers is insufficient for rare-event forecasting: while aggregate accuracy appears strong, the models fail to capture the high-impact, low-frequency hazard events that are most consequential for port operations.

B. Physics Aware PORTEX and Interpretability

Incorporating domain knowledge through a weighted linear PORTEX model resulted in the model achieving an AUC of 0.833. This value represents an 11% increase over baseline methods, and reveals how this inclusion substantially enhances predictive performance by explicitly encoding physically and operationally relevant features into the model architecture based on domain-specific knowledge about how storms impact ports. Furthermore, Figure 3 reveals that Vulnerability (+17.61) and Hazard (+3.21) are the primary drivers of risk. These findings align with operational intuition that fragile infrastructure and storm intensity dominate port disruption outcomes. Additionally, the negative coefficient associated with Proximity (-0.62) suggests that storms closer to ports exhibit more predictable trajectories, which is a subtle finding that emphasizes the importance of integrating physical context rather than relying solely on storm magnitude.

Moreover, the stability of coefficients across LOSO cross-validation folds distinguishes PORTEX from tree-based ensembles, whose feature rankings vary significantly under small changes in the training set. This stability is important as it reveals that the model’s learned representation is invariant to storm-specific disturbances, providing reliable interpretability allowing for better operational decision-making.

C. Uncertainty Calibration with Streaming Conformal Prediction

In the context of sparse high-risk hazard events, calibrated uncertainty estimates are essential for operational deployment. Integrating rolling-window conformal prediction with PORTEX and logistic regression preserved predictive performance,

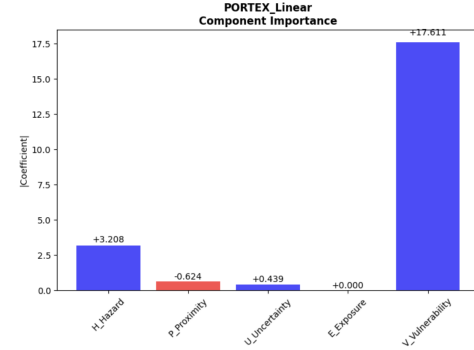


Fig. 4. PORTEX Feature Importance

achieving an AUC of 0.74–0.75 while also providing empirically conservative coverage. For a nominal miscoverage rate of $\alpha = 0.05$, observed coverage reached 99.4%, and for $\alpha = 0.10$, coverage reached 99.2%, demonstrating that the conformal framework maintains distribution-free validity even under inter-storm shifts.

Prediction efficiency remained high, with average prediction set sizes ranging from 1.0002 to 1.0020, effectively corresponding to single-point predictions, ensuring that coverage guarantees were achieved without inflating prediction sets. These calibrated intervals further enable operators to balance false alarms with missed detections using dynamic threshold adjustment, offering an operational advantage over conventional machine learning models that provide point estimates without reliable error bounds.

D. Sequential Modeling Performance

Lightweight sequential models were evaluated to capture temporal dependencies in advisory streams, a capability rarely explored in the context of few-shot, rare-event port disruption prediction. Unlike static models, sequential architectures, such as the Compact GRU and Conv1D, can explicitly model the evolution of storm characteristics over time, including gradual intensification, trajectory shifts, and accumulating operational impact. This temporal modeling is crucial in a streaming setting, where risk estimates must be updated dynamically with each advisory. Table 1 reveals that Compact GRU achieved an AUC of 0.797 (+6.3 percent over baselines), while Compact Conv1D achieved 0.757. These compact architectures (hidden dimensions ;16) are designed to maintain real-time infer-

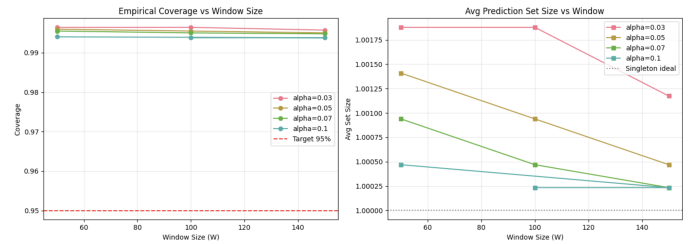


Fig. 5. Conformal Prediction Uncertainty Calibration

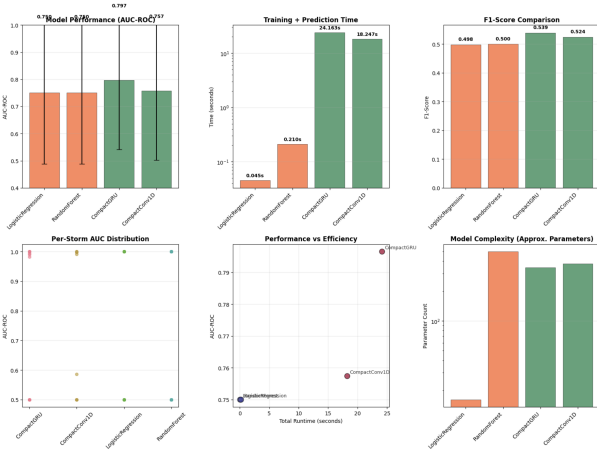


Fig. 6. Sequential Modeling Performance Analysis

ence feasibility, ensuring that temporal dependencies can be leveraged without prohibitive computational overhead. These findings demonstrate that while physics-informed features dominate predictive performance, sequential models provide an additional layer of temporal context, enabling more robust updates to risk estimates as storms evolve and the utility of integrating sequential modeling as a complementary component.

CONCLUSION

This paper proposes FS-PREM, a framework to predict rare port disruption events from streaming hurricane advisories. By integrating physics-aware PORTEX features along with streaming conformal prediction, FS-PREM achieved an 11% AUC improvement over baseline methods while retaining > 99% uncertainty coverage under inter-storm distributional shifts.

To drive this improvement, coefficient analysis indicates that among the PORTEX features, Vulnerability (+17.61) and Hazard (+3.21) are the primary risk drivers, suggesting that fragile infrastructure and storm intensity dominate disruption outcomes. Furthermore, lightweight sequential models such as the Compact GRU, which achieved an AUC of 0.797, enable the model to provide real-time inference capabilities on streaming advisory data. Ultimately, these findings demonstrate the robust foundation for port risk assessment that physics-informed feature engineering, in tandem with calibrated uncertainty qualification, provides.

LIMITATIONS AND FUTURE WORK

While the FS-PREM framework establishes a strong foundation for forecasting rare port disruption events, certain constraints and assumptions merit discussion:

First, the PORTEX index currently relies on fixed, heuristic weights for hazard, proximity, uncertainty, exposure, and vulnerability. Although interpretable, this static approach may underrepresent nonlinear storm–port dynamics. Future work could allow weights to be adaptively learned from data while maintaining domain constraints.

Second, the conformal prediction framework assumes exchangeability across advisories, which may not hold under rapidly shifting storm trajectories. Future research should explore methods such as non-exchangeable conformal approaches or Bayesian calibration techniques to relax this assumption.

Third, the dataset is limited to Gulf Coast hurricanes post-COVID, restricting generalizability to other hazards and regions. Expanding FS-PREM to include typhoons, floods, or wildfires would allow testing model portability and support development of transfer learning approaches across event types.

V. ACKNOWLEDGMENTS

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